Type 2 diabetes: an overview
1. What's in this GEMS and what's new
Based on NICE T2DM/CKD guidance (NG28 2022 & 203, 2021)

Red Whale
GEMS
Guidelines & Evidence Made Simple

Diabetes is a complex interplay of many factors, all of which contribute to the development of the disease and impact on management. Here, we have tried to summarise care, focusing particularly on glycaemic control because that is what many of us clinicians find challenging.

THROUGHOUT CARE

Remember the most important part of care...
Lifestyle! Lifestyle! Lifestyle!

Remember LIFESTYLE!
See section 3

What's going on in T2DM?
Macro & micro pathology Section 2

Monitor and manage
Psychological wellbeing
Diet and lifestyle
Peripheral/autonomic neuropathy
Feet
Eyes
HbA1c
Cholesterol
BP
Renal health

See section 8

Avoid therapeutic inertia
Failure to appropriately intensify/de-intensify therapy

In practice? Example of a practice protocol See section 9

What's NEW! in NICE 2022 guidance?
It's all about gliflozin use...

1st **First line with metformin** in those at increased CV risk/known CVD

For cardioprotection if at any stage develops CVD or risk factors for CVD. Use instead of, or in addition to, existing drugs

For renoprotection in CKD

GLYCAEMIC CONTROL

Lifestyle central to good diabetic control
Lifestyle!

Continually review understanding of:
- Diabetes as a cardiometabolic condition
- Lifestyle interventions that improve outcomes:
  - Diet, weight loss, activity, smoking cessation

See section 3

Agree and work towards targets for glycaemic control (and BP/cholesterol) based on comorbidities, life expectancy and patient preference. *Lifestyle change benefits all targets and much more!*
Review barriers to change, especially deprivation

See section 4

If HbA1c remains above target, review:
- Target *(is it still appropriate?)*
- Lifestyle
- Compliance

AND START DRUG THERAPY

FIRST-LINE DRUG THERAPY
(unless these drugs are contraindicated)

See section 5

Metformin and NEW! Gliflozin or Metformin alone

Decision to start dual or monotherapy depends on CV risk
Use ALONGSIDE lifestyle

If HbA1c remains above target, review:
- Target *(is it still appropriate?)*
- Lifestyle
- Compliance

AND INTENSIFY DRUG THERAPY

INTENSIFICATION OF DRUG THERAPY
ADD another drug *or* SWITCH to alternative drug

Which drug to use based on:
- Patient preference and comorbidities
- Effectiveness and harms/side-effects of drug

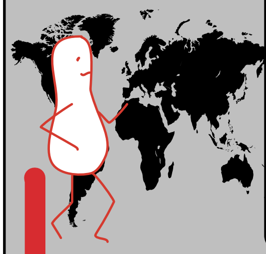
See sections 6 & 7

If intensification doesn't achieve target, review target, lifestyle, compliance and, if appropriate, intensify further

2. What's going on in T2DM?

Lancet 2022;400:1803

Macro level



- Obesogenic environment
- Diet
  - Chronic positive energy balance (calories in > calories burnt)
  - Eating the wrong proportions of each food group
- Sedentary lifestyle and insufficient aerobic activity

And in the developing world:

- Urbanisation
- Rapid economic development

Relative impact

Policy

Individual

Individuals have some impact, but population-level interventions are needed to change the environment

Complex cardiorenal metabolic disorder

Glucose and lipid dysregulation affect vascular system, resulting in microvascular disease (organ dysfunction) and macrovascular disease (strokes, heart disease)

Micro level

Insulin resistance

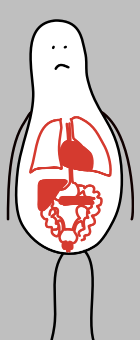
- Especially in liver, skeletal muscles and adipose tissue

Pancreatic dysfunction

- Reduction in mass and function of beta cells (impaired insulin release)
- Defects in alpha cells (increasing glucagon production)
- Islet amyloid polypeptide (the hormone related to insulin and glucagon secretion) deposition in pancreas, which may lead to destruction of beta cells

Gastrointestinal dysfunction

- Abnormal stomach emptying
- Increased glucose absorption
- Dysbiosis of gut microbiota (loss of beneficial bacteria, overgrowth of potentially pathogenic bacteria, loss of overall bacterial diversity)
- Reduced effect from incretins (augment insulin secretion, inhibit glucagon release)



Renal dysfunction

- Increase glucose reabsorption, in part through SGLT2 over-expression
- Inflammation and fibrosis

Brain dysfunction

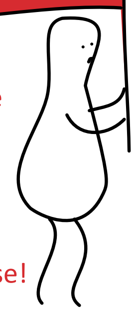
- Impaired appetite regulation
- Increased sympathetic tone
- Reduced dopamine activity

Chronic inflammation

Immune dysfunction

Relative impact

Drugs have impacts across one or two of these



Lifestyle change has benefits across ALL these!

3. Lifestyle: weight and activity

From ADA and European Association for the Study of Diabetes (EASD) (Diabetes Care 2022;45:2753)

How to encourage change?
See our article *Motivating behaviour change*

Key elements of education, to increase understanding of:

- Diabetes as a cardiometabolic condition
- Lifestyle interventions that improve outcomes (weight loss, nutrition, activity, smoking cessation)
- Medication (what it is for, how to take it, benefits, adverse effects)

Weight loss

Progressive benefits seen with increasing weight loss (from 5–15% loss). Overall, ≥10% may be required to see cardiovascular and mortality benefits or reduction in complications (e.g. steatohepatitis seen in NAFLD)

NUTRITION **Aim: healthy habits that are feasible, sustainable and do not remove the pleasure of eating.**

- **Bottom line: any (sensible!) diet resulting in energy deficit (calorie intake < calories burnt) beneficial.**
- Greatest glycaemic control benefits seen with Mediterranean and low-carb diets. Mediterranean diet has best evidence in terms of maintaining weight loss. Intermittent time-restricted fasting (e.g. fasting 18h a day) and 5:2 beneficial but no more so than other approaches.

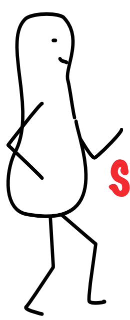
DRUGS **Orlistat can be prescribed in primary care, and liraglutide and semaglutide through NHS weight management services.** Semaglutide and tirzepatide are more effective than older therapies. Semaglutide is a GLP-1 mimetic, available as weekly injection or daily tablet. Tirzepatide acts on the GLP-1 pathway to activate both GLP-1 and GIP receptors; weekly injection. More on this in *Obesity: pharmacological management* article.

SURGERY **Significantly more effective at getting people into remission than any medical therapy, but has beneficial metabolic and glycaemic outcomes even if remission not achieved.**
Approx. 40% achieve remission but varies by type of surgery. More effective within a few years of diagnosis (significant reduction in chance of remission if had diabetes longer than 5–8y).
NICE criteria for bariatric surgery (for those diagnosed in past 10y, NICE CG189, updated 2023):

- Offer if BMI ≥35
- Consider if BMI 30–34.9

If of Asian, South Asian, Chinese, Middle Eastern, Black African or African-Caribbean family background, consider referral at BMI ≥27.5

Activity



SIT LESS

STEP MORE

SWEAT MORE

Any increase in physical activity is good! Benefit of activity is a continuum: even small changes beneficial. Activity also improves mood and quality of life.

500 extra steps/day associated with 2–9% fall in CV and all-cause mortality (most data from relatively inactive populations). Least active get most benefit.

Regular aerobic exercise improves:

- Glycaemic control
- Daytime hyperglycaemia
- CV endpoints (BP, lipids)

Glycaemic benefits maximised if:

- Activity taken post-prandially
- Exercise ≥45mins

STRENGTHEN MUSCLES

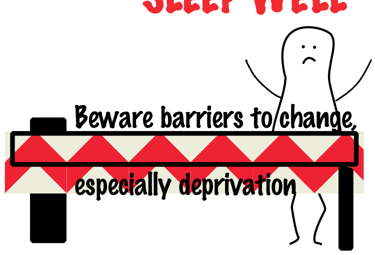
Strength/resistance training improves:

- Blood glucose levels
- Flexibility/balance: important because diabetes is an accelerator of biological ageing, leading to sarcopaenia and frailty at a younger age.

SLEEP WELL

Poor sleep worsens glycaemic control and increases risk of obesity. Half of people with T2DM have sleep apnoea, and severity of this is associated with worsening glycaemic control. *Something to specifically ask about?* (Good sleep quality also improves BP, lipid profile and QOL).
The following have a negative impact on HbA1c:

- Quantity of sleep: too long or too short (<6h or >8h)
- Quality of sleep: irregular sleep (e.g. shifts, sleep apnoea, restless legs)
- Owls may be more at risk of inactivity/poor glycaemic control than larks



4. NICE on drugs for glycaemic control

NICE NG28, updated 2022. Many drugs restricted even in mild renal impairment: check BNF. Not an exhaustive list; for more details and references, see our diabetes articles

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LIFESTYLE is first-line therapy and important THROUGHOUT CARE

Set personalised targets

Set targets for HbA1c/cholesterol/BP: this helps guide intensity and choice of treatment.

Set these targets based on: **patient preference** and **frailty.**

Remember, it isn't all about glycaemic control: ensure good BP/cholesterol control too!

Best wins for **macro**vascular protection. Probably: 1. BP control 2. Cholesterol-lowering 3. Glycaemic control.

Less clear for **micro**vascular protection. Probably: 1. BP control 2. Glycaemic control.

Patient preference

Patient may be keen to take additional medication to achieve tighter control, or may be reluctant to do so because of the burden of polypharmacy or other reasons.

Frailty

Long life expectancy
Minimal comorbidities
Minimal risks from polypharmacy

Aim for tighter control of HbA1c/BP/cholesterol to reduce risk of long-term complications.

Frail, limited life expectancy
Multiple comorbidities
Risks from polypharmacy

Long-term outcomes less important? Focus on quality of life: don't make treatment over-burdensome. **Targets may be relaxed.**

NICE targets
(suggested targets, personalise as appropriate)

Lifestyle alone Single drug that doesn't cause hypos	48mmol/mol or 6.5%
On hypo-inducing drug (sulphonylureas, insulins) On more than one drug (recommendation predates advice to use dual therapy first line in some, but was not amended in 2022 update)	53mmol/mol or 7%

If HbA1c rises above target → **Review:**
- Target *(is it still appropriate?)*
- Lifestyle
- Compliance

→ **Start or intensify drug therapy** (Lifestyle!)

! Avoid therapeutic inertia: failure to appropriately intensify (or deintensify) therapy !

Personalisation of care

Diabetes is a complex interplay of many factors, all of which contribute to the development of the disease and impact on management. *Many of these factors are out of the control of the individual.*

BIOLOGICAL FACTORS

Diabetes is a heterogeneous condition with individual variations in:
- Degrees of obesity
- Degree of insulin resistance
- Tendency to complications

PSYCHOSOCIAL FACTORS
- Behavioural factors
- Emotional factors
- Social factors
- Environment

SOCIAL DETERMINANTS OF HEALTH
- Socioeconomic status (education, income, occupation)
- Living and working conditions, culture
- Sociopolitical context/government policies

5. NICE on first-line drug therapy

Many drugs restricted even in mild renal impairment: check BNF

Not an exhaustive list; for more details and references, see our diabetes articles

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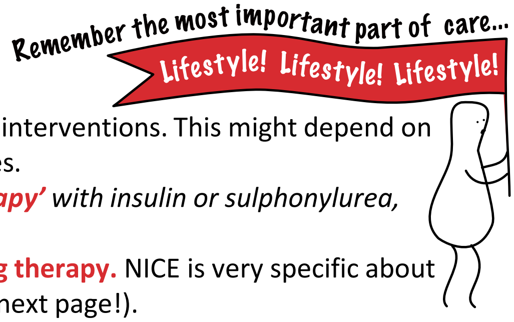


Starting first-line drug therapy

LIFESTYLE is first-line therapy and important THROUGHOUT CARE.
NICE does not recommend a particular period of time to try lifestyle interventions. This might depend on patient-specific factors such as motivation to change or comorbidities.
If SYMPTOMATIC HYPERGLYCAEMIA present, consider 'rescue therapy' *with insulin or sulphonylurea, then review treatments once glycaemic control achieved.*
If personalised targets are not met, NICE recommends starting drug therapy. NICE is very specific about first-line therapy (and much less so about subsequent options – see next page!).

NICE recommends:

FIRST-LINE DRUG THERAPY

(unless these drugs are contraindicated)

Metformin and Gliflozin NEW! or Metformin alone

Decision to start dual or monotherapy depends on CV risk.
Use ALONGSIDE lifestyle.

Consider CV risk

Which first-line drug therapy?

Metformin for all (unless contraindicated).

NEW! If increased cardiovascular risk, add gliflozin once stabilised on metformin (see below).

Atherosclerotic CVD (IHD/ischaemic stroke/TIA/PAD) **or chronic heart failure**

Metformin **AND OFFER** Gliflozin

High risk of CVD
- **≥40y and QRISK >10%**
- **<40y and ≥1 CV risk factors:**
  - Smoker
  - Obese
  - Hypertension
  - Dyslipidaemia
  - Premature CVD in first-degree relative

Metformin **AND <u>CONSIDER</u>** Gliflozin

Everyone else

Metformin alone

Which gliflozin?

NICE recommended using a gliflozin "with proven cardiovascular benefit". There are no head-to-head trials but, through statistical analysis, NICE concluded that:
- It is unclear if there are real differences in cardiovascular benefits between the different gliflozins.
- There was greater uncertainty around the CV benefits of ertugliflozin than the other gliflozins.
- Note that dapagliflozin should be used if starting a gliflozin for renoprotection in CKD (see section 7).

Starting metformin and gliflozin as dual therapy

Start metformin (NICE suggests titrating up over several weeks to minimise GI side-effects), THEN add gliflozin once metformin tolerability established. **Before starting gliflozins, check:**
- Not at increased risk of DKA, e.g. previous DKA or on very-low carbohydrate or ketogenic diet.
- Not pregnant/planning to get pregnant/breastfeeding.
- Renal function acceptable and renal function can be monitored (it doesn't specify how often).

COUNSEL about SICK DAY RULES and DKA RISK: stop gliflozins + metformin in acute illness (e.g. fever, D&V).

If metformin is contraindicated/not tolerated

If metformin contraindicated/not tolerated, NICE recommends:
- **Known CVD/CCF:** <u>offer</u> gliflozin as monotherapy.
- **High risk of CVD**: <u>consider</u> gliflozin as monotherapy.
- **Low CVD risk**: choose from any oral therapy (gliptin, pioglitazone, sulphonylurea or gliflozin (NICE has specific recommendations on gliflozin use in those <u>without</u> increased CV risk – see our article on glycaemic control for details)).

6. Intensification: which drug when?

Many drugs restricted even in mild renal impairment: check BNF

Not an exhaustive list; for more details and references, see our diabetes articles

Red Whale GEMS Guidelines & Evidence Made Simple

Offer intensification if personalised targets not met.

NICE recommends:

INTENSIFICATION OF DRUG THERAPY

ADD another drug *or* SWITCH to alternative drug

If intensification doesn't achieve target, review target, lifestyle, compliance and, if appropriate, intensify further

If SYMPTOMATIC HYPERGLYCAEMIA develops, consider insulin or sulphonylurea as 'rescue therapy', then review treatments when glycaemic control achieved.

So, how do you choose which drug to add or switch?

NICE suggests:

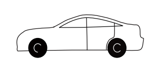
- Any oral therapy (gliflozins in line with NICE guidance)
- GLP-1 only in line with NICE guidance (outlined overleaf)
- Insulin (if insufficient control on 2 oral drugs)
- Consider a gliflozin if, at any stage, someone develops CVD, CHF or increased CV risk **NEW!**

Based on NICE guidance and current evidence, we suggest considering **patient preference** and **comorbidities**

Patient preference

May be affected by how drugs taken, side-effects and monitoring requirements

Comorbidities may guide treatment

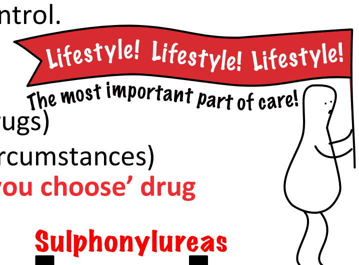
Want **cardiovascular protection? Consider…** **GLIFLOZINS** (Renoprotection? See section 7) **AVOID** pioglitazone in heart failure 	Significant **renal impairment? Consider…** **INSULINS** **PIOGLITAZONE** **some GLIPTINS (linagliptin)** 
Very **obese? Consider…** **GLP-1s** **GLIFLOZINS** 	Need to **AVOID significant hypoglycaemia?** **AVOID insulins** **AVOID sulphonylureas** 

And, remember, it isn't all about glycaemic control: ensure good BP/cholesterol control too!

Best wins for **macro**vascular protection. Probably: 1. BP control 2. Cholesterol-lowering 3. Glycaemic control.

Less clear for **micro**vascular protection. Probably: 1. BP control 2. Glycaemic control.

Still not sure?



Think of the drugs in 2 groups: *'Metformin's mates'* (commonly used as step-up drugs)

'Use because you choose' drugs (used in specific circumstances)

Use one of 'Met's mates' UNLESS there is a good reason to pick a 'use because you choose' drug

The key pros and cons of *'Met's mates'* are (more details overleaf):

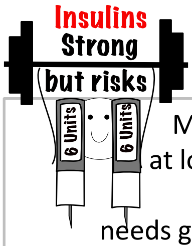
Well-tolerated
May result in weight loss
Cardioprotective
Renoprotective if CKD
Beware risk of DKA!

Low hypo risk and weight neutral
BUT
least effective to lower HbA1c

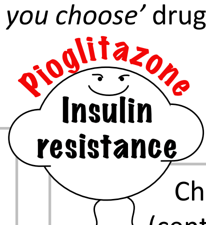
Very effective at lowering HbA1c
BUT
hypo risk AND weight gain

Real-world, UK-based observational data shows that gliflozins outperformed gliptins and sulphonylureas for: glycaemic control, BMI, BP, heart failure and renal outcomes (but, interestingly, not MACE) (see main article, BMJ 2024;385:e077097)

The reasons to pick a *'use because you choose'* drug are:



Most effective at lowering HbA1c
BUT
needs good understanding AND significant hypo risk



Choose if INSULIN RESISTANCE (central obesity, high insulin doses)
BUT contraindicated in several conditions (see overleaf)

NICE reserves for obesity

7. Key pros & cons of diabetes drugs

Many drugs restricted even in mild renal impairment: check BNF

Not an exhaustive list; for more details and references, see our diabetes articles

Red Whale GEMS Guidelines & Evidence Made Simple

When choosing a drug, considering the following can also help: how effective is the drug? Does it increase the risk of hypos? What impact does this drug have on weight? If my patient has renal impairment, is this drug safe? What are the harms and cautions for this drug, and how relevant are these to the patient before me? If all other things are equal, is this a cost-effective choice? This table should help answer these! (based on SIGN 154, 2017)

	ADVANTAGES	DISADVANTAGES
First line		
METFORMIN for all unless contraindicated	Cardioprotective Low risk of hypos May help weight loss	Renally excreted: accumulates in AKI/CKD SPC: eGFR<30: do not use eGFR 30–44: max. 1g/d, eGFR 45–59: max. 2g/d SICK DAY RULES APPLY! Consider stopping in acute illness/risk of dehydration
GLIFLOZINS (SGLT2i) If CVD/CCF/high CV risk	Low risk of hypos May result in weight loss Well-tolerated Cardioprotective in those with and without CVD Renoprotection (mainly in those with CKD)	'Euglycaemic' DKA RISK! SICK DAY RULES APPLY! Consider stopping in acute illness/risk of dehydration Care in renal impairment Increased risk of genital infections, possible increased risk of amputation, esp. if dehydration
'Met's mates': *when intensifying treatment, these drugs would normally be used unless there was good reason to pick a 'use because you choose' drug*		
GLICLAZIDE	One of the MOST EFFECTIVE drugs to lower HbA1c	High risk of hypos (esp. in renal impairment) Causes weight gain
GLIPTINS	Low risk of hypos Weight neutral	One of the LEAST effective drugs to lower HbA1c Increased risk of cholecystitis (see main article)
GLIFLOZINS (SGLT2i)	*(If not already taking)* Use if atherosclerotic CVD or CHF or high CV risk. If no increased CV risk, NICE has specific recommendations on gliflozin use and which drugs they can be used alongside: see article on glycaemic control for details as it varies from gliflozin to gliflozin. Also used in CKD: see section 7.	
'Use because you choose' drugs: *pick one of these if you have a specific reason to*		
INSULINS	Reason to choose: MOST EFFECTIVE HbA1c LOWERING Safe in renal impairment *NICE guidance: consider insulin if insufficient control on 2 oral drugs*	Hypo risk! SICK DAY RULES APPLY! Good understanding needed Given by injection Causes weight gain
PIOGLITAZONE	Reason to choose: INSULIN RESISTANCE (central obesity, high insulin requirement) Low risk of hypos Safe in renal impairment	Causes weight gain Increased risk of heart failure Contraindicated in heart failure/bladder cancer Concern re fractures. *Caution in elderly!*
GLP-1s (e.g. liraglutide) *See GEMS on GLP-1s and tirzepatide for more on this*	Reason to choose: NICE RESERVES FOR OBESITY Low risk of hypos Often result in weight loss Cardioprotective (most notably in those with CVD) Renoprotection mainly in those with CKD	Expensive All injected except semaglutide (oral or sc) *MHRA (2024) side-effects:* GI side-effects (nausea, vomiting, diarrhoea, constipation) common (>1 in 10), but can lead to severe dehydration needing hospitalisation Rare but serious: pancreatitis, gallstones

NICE CRITERIA for use of GLP-1s and tirzepatide: US/European guidance offers GLP-1s as a first-line option if CVD/at high risk of CVD. **NICE limits use to those with obesity if triple oral therapy not effective/tolerated or contraindicated.**

- BMI ≥35 (lower if black/Asian/minority group) AND specific obesity-related medical problem *(BMI ≥32.5 if Asian, South Asian, Chinese, Middle Eastern, Black African or African-Caribbean).*
- Lower BMI if weight loss would help a significant obesity-related comorbidity OR occupation makes insulin unsuitable.

For GLP-1 (not tirzepatide): continue only if 11mmol/mol (1%) reduction in HbA1c AND weight loss of 3% in first 6m.

8. BP/lipids/CKD in T2DM

Many drugs restricted even in mild renal impairment: check BNF

Not an exhaustive list; for more details and references, see our diabetes articles

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This is a SUMMARY of the NICE guidance relevant to BP/cholesterol and CKD. For detailed information, see the relevant articles.

Remember the most important part of care... Lifestyle! Lifestyle! Lifestyle!

Blood pressure control (from NICE NG136, 2019)

BP targets as per general population, but use ACEi/ARB first line regardless of age/ethnicity.

Aim for clinic BP of...

<140/90 (<150/90 if ≥80y)

(If CKD, see below)

Drug therapy for BP

Use ACEi/ARB first line for all (unless contraindicated)

Lipids and statins (from NICE 2023, NG238)

Manage as per general population, which means...

Primary prevention	**25–85y**	**If QRISK3 ≥10%: offer atorvastatin 20mg.** *If QRISK3 <10%: statin may be used if patient chooses or there is concern risk is underestimated (e.g. recently stopped smoking).*
	≥85y	**High risk based on age alone so do not do QRISK.** ***CONSIDER*** **atorvastatin 20mg** (taking account of frailty, comorbidities, polypharmacy).
Secondary prevention (known CVD)		**OFFER atorvastatin 80mg** (20mg if CKD). • **If target NOT met, consider adding non-statin lipid lowerers.** • **Even if target met, *CONSIDER* adding ezetimibe to statin to further reduce risk.**

Target in...

PRIMARY PREVENTION

>40% reduction in non-HDL cholesterol

SECONDARY PREVENTION

LDL ≤2 or non-HDL ≤2.6

Managing CKD in type 2 diabetes (from NICE NG203, updated 2021)

CKD defined as any 1 of:

- *ACR ≥3 (with any eGFR)*
- *eGFR <60 on 2 occasions at least 3 months apart*
- *Structural kidney disease (e.g. polycystic kidney disease), even if normal eGFR*

BP control in CKD

Target clinic BP...

CKD and ACR <70: <140/90

CKD and ACR ≥70: <130/80

Drug therapy for BP

Use ACEi/ARB first line for all (unless contraindicated)

If proteinuria (with or without hypertension)

NEW!

If ACR 3–30

ARB/ACEi

AND *CONSIDER* Gliflozin

If ACR >30

ARB/ACEi

AND *OFFER* Gliflozin

(Gliflozin must be licensed for use in CKD – currently dapagliflozin and empagliflozin)

Dapagliflozin and empagliflozin can be used outside the above guidance if patient meets certain criteria (see CKD article for more info) (NICE 2022, TA775, 2023 TA942).

Cholesterol in CKD

Drug therapy for cholesterol

Offer 20mg atorvastatin for primary AND secondary prevention.

Increase dose only if cholesterol targets (see above) not met,

BUT if eGFR <30, increase dose ONLY on specialist advice.

9. Stepping-up drugs in T2 diabetes
Mill Stream Surgery Protocol

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Based on our practical application of the Red Whale *An approach to the diabetic drugs: GEMS*, NICE and local guidance. At every stage: think LIFESTYLE first & STOP ineffective drugs. **Lifestyle changes give the biggest win across all diabetic/CV outcomes**. If **rescue therapy** (rapid glycaemic control) needed at any stage, e.g. symptomatic hyperglycaemia, consider gliclazide or insulin.

High risk of CVD =

- ≥40y and QRISK >10%
- <40y and ≥1 CV risk factors:
  - Smoker
  - Obese
  - Hypertension
  - Dyslipidaemia
  - Premature CVD in 1st-deg relative

Which first-line therapy?

Metformin unless contraindicated **AND, if CVD/high risk CVD,** Gliflozin unless contraindicated (must understand sick day rules/risk of DKA)

Intensification needed?

If not hitting targets, review: targets (still appropriate?), lifestyle, compliance, and intensify treatment. Choice of treatment based on patient preference and comorbidities. Consider the following:

CVD or high risk of CVD?	Very obese?	Renal failure?	Need to AVOID significant hypos?
GLIFLOZINS (but check BNF as some can't be used in renal impairment) **AVOID pioglitazone in heart failure**	**GLP-1s** **GLIFLOZINS**	**INSULINS** **PIOGLITAZONE** **some GLIPTINS (linagliptin)**	(e.g. drivers) **AVOID insulins** **AVOID sulphonylureas**

If still not sure…

Use 'Met's mates' *(Glif, Glip and Glic)….*

- **Gliflozin**: Use if CVD/high risk of CVD. Renoprotective if CKD (minimal benefit if no CKD). Well-tolerated. May result in weight loss. Beware risk of DKA!
- **Gliptin**: Low hypo risk. Weight neutral but least effective to lower HbA1c
- **Gliclazide**: Very effective at lowering HbA1c but hypo risk and weight gain

….unless you 'use because you choose' (GLP-1, pio, insulins)

- **Significant obesity** → **GLP-1**
- **Insulin resistance** (central obesity, needing high insulin doses) → **Pioglitazone**
- **Insulin required** (patient choice, very effective, renal failure) → **Insulins**

Still not meeting targets? Intensify further

Drugs (limited info given here: always check the BNF before prescribing)	
Metformin	Initially 500mg with breakfast. Max. 3g/day in 2–3 doses, but above 2g/day unlikely to see extra benefit. Use in everyone unless very good contraindication (CV benefits). **Sick day rules apply! Stop in acute illness (renally excreted). eGFR <30: do not use. eGFR 30–44: max. dose 1g/d. eGFR 45–59: max. 2g/d.**
Gliclazide	Initially 40–80mg daily with breakfast. Max. 320mg (above 160mg split into twice-daily dose). **Risk of hypos!** And this risk is even higher in renal impairment.
Gliptin	Alogliptin (Vipidia) 25mg once daily. On starting: reduce gliclazide/insulin dose (reduces risk of hypos). **CrCl <30: 6.25mg daily. CrCl 30–50: 12.5mg daily.**
Gliflozin	Dapagliflozin (Forxiga) 10mg once daily. Do not initiate if eGFR <15. Glucose-lowering less effective if eGFR <45. On starting: reduce gliclazide/insulin dose (to reduce risk of hypos). **DKA RISK at relatively low blood glucose! Must understand sick day rules. Check ketones/stop if unwell!**
Pioglitazone	**Reserved for insulin resistance** (central obesity, high insulin need). Caution in elderly (risks of heart failure, bladder cancer, fractures). Initially 15–30mg daily, maximum 45mg once daily. Safe in renal impairment.
GLP-1s Tirzepatide (Mounjaro)	**Reserved for obesity.** NICE: BMI ≥35 **and** weight-related comorbidity or BMI <35 if can't use insulin because of occupation/weight loss desirable. BMI threshold 2.5 lower in ethnic minorities. Beware GI side-effects, dehydration, pancreatitis. Caution in renal impairment. DKA risk if insulin stopped suddenly. (MHRA)